

PRODUCER THEORY

The Three Stages of Production

The production process in the short run is divided into three stages, based on the relationship between Total Physical Product (TPP), Average Physical Product (APP), and Marginal Physical Product (MPP).

Stage 1: Increasing Returns

In this stage, MPP is greater than APP, meaning additional inputs increase efficiency.

APP is rising because each additional unit of input (e.g., labor) significantly increases output.

TPP increases at an increasing rate.

Firms should keep adding inputs because productivity is still improving.

Example: A farm hiring more workers experiences rapid growth in output because workers use the land more efficiently.

Stage 2: Diminishing Returns

MPP starts to decline, but it is still positive.

APP reaches its maximum and then starts to fall.

TPP increases but at a decreasing rate.

This is the ideal stage for production, as firms get the most efficient use of inputs.

Example: The farm hires more workers, but each extra worker contributes less additional output due to limited space.

Stage 3: Negative Returns

MPP becomes negative, meaning adding more inputs decreases output.

TPP starts to decline.

APP continues to fall.

Firms should stop adding inputs as they are now overcrowding and becoming less productive.

Example: Too many workers on a small farm get in each other's way, reducing overall efficiency.

A farmer should stop growing (adding more inputs like labor or fertilizer) at Stage 3 (Negative Returns) because at this point:

- Marginal Product (MPP) is negative, meaning adding more inputs reduces total output.
- Total Physical Product (TPP) starts to decline, meaning the farm is producing less, not more despite using more resources.
- Resources are wasted, as extra labor or fertilizer becomes counterproductive (e.g., overcrowding reduces efficiency).

Ideal Stopping Point:

The farmer should operate in Stage 2, where MPP is still positive but decreasing.

The best point to stop adding inputs is when $MPP = 0$, which is the maximum TPP before it starts declining.

A farmer should not stop at Stage 1 because they are not yet fully utilizing resources efficiently. Here's why:

1. APP is still rising → Each additional input (e.g., worker or fertilizer) is increasing productivity.
2. $MPP > APP$, meaning adding more inputs still improves efficiency.
3. TPP is increasing at an increasing rate, so the farmer can produce much more output by continuing into Stage 2.

Why Stop in Stage 2 Instead?

Stage 2 is the most efficient stage because MPP is still positive, and output is growing, but at a slower rate.

Stopping in Stage 1 means missing out on potential higher yields before reaching the optimal production level.

Firms and farmers maximize efficiency by continuing production until $MPP = 0$ (end of Stage 2).

Graph-Related Questions (5)

1. What does the Total Physical Product (TPP) graph show?

The TPP graph shows the relationship between the number of workers (Lb) and the total wheat output (tonnes per year). It illustrates three stages of production:

Stage 1: Increasing returns (TPP rises sharply).

Stage 2: Diminishing returns (TPP rises but at a decreasing rate).

Stage 3: Negative returns (TPP declines).

2. At what point does the Marginal Physical Product (MPP) reach zero, and what does it indicate?

MPP reaches zero at 7 workers. This indicates that total output (TPP) is at its maximum. Adding more workers beyond this point will result in negative returns.

3. How does the Average Physical Product (APP) behave in relation to MPP on the graph?

When $MPP > APP$, APP is rising.

When $MPP < APP$, APP starts to fall.

MPP = APP at the peak of APP, meaning the highest efficiency per worker.

4. What does the drop in TPP after 7 workers mean for a farmer?

It means the farm has too many workers, leading to inefficiencies. This is Stage 3 (Negative Returns), where extra labor lowers output rather than increasing it. The farmer should not hire more workers beyond 7.

5. Based on the table, calculate APP for 4 workers if TPP is 36.

$$APP = \{TPP\} / \{Lb\} = \{36\} / \{4\} = 9$$

General Theory Questions (5)

6. What is the Law of Diminishing Returns?

It states that as more units of a variable input (e.g., labor) are added to a fixed input (e.g., land), output will initially increase at an increasing rate, then at a decreasing rate, and eventually start to fall.

7. Differentiate between the short run and the long run in production.

Short run: At least one input is fixed (e.g., land, buildings).

Long run: All inputs can be varied (firms can expand factories, buy more land, etc.).

8. Why should firms operate in Stage 2 of production?

MPP is positive, so adding more inputs still increases output.

TPP is at its highest before declining.

APP is maximized, meaning efficiency per worker is highest.

9. What happens when MPP is greater than APP?

APP is rising, meaning each additional input increases efficiency. This happens in Stage 1 of production.

10. Define and explain the production function.

The production function shows how output (TPP) depends on different inputs (capital, labor, materials). It follows the equation:

$$TPP = f(K, L, M)$$